

Helyasadat Hashemi Aghdam

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Department of Computer
Engineering
Bilkent University
Çankaya, Ankara
Turkey

Education

- 2024 – 2027 **M.Sc. in Computer Engineering** (expected 2027)
Bilkent University, Ankara, Turkey
Thesis: in progress
Advisor: [Dr. A. Ercüment Çiçek](#) – **CGPA:** 3.80/4.00
- 2020 – 2024 **B.Sc. in Computer Engineering**
University of Tabriz, Tabriz, Iran
Final Project: Variational contrastive self-supervised learning for disentangled representations
GPA: 18.27/20 (3.79/4.00)

Publications

Peer-Reviewed Journal Articles

- 2025 “[RNAtranslator: Modeling protein-conditional RNA design as sequence-to-sequence natural language translation](#)”, Sobhan Shukueian Tabrizi, Sina Barazandeh, **Helyasadat Hashemi Aghdam**, A. Ercüment Çiçek. *PLOS Computational Biology*, 21(10), e1013541. (Presented at **ISMB/ECCB 2025**)

Preprints

- 2025 “[RNA-X: Modeling RNA interactions to design binder RNA and simultaneously target multiple molecules of different types](#)”, Sobhan Shukueian Tabrizi, **Helyasadat Hashemi Aghdam**, A. Ercüment Çiçek. *bioRxiv* 2025.11.24.690191. (Presented at **RECOMB-RSG 2026**)

Manuscripts Submitted

- 2026 “From Biased Structural Benchmarks to Generalizable RNA-Protein Interface Prediction”, **Helyasadat Hashemi Aghdam**, Sobhan Shukueian Tabrizi, A. Ercüment Çiçek. (Submitted, **NeurIPS 2026**)
- 2026 “Cross-Target Transfer Enables Small-Molecule-Binding RNA Generation”, **Helyasadat Hashemi Aghdam**, Orhun Ege Çelik, Sobhan Shukueian Tabrizi, A. Ercüment Çiçek. (Submitted, **NeurIPS 2026**)

Skills

- **Programming:** Python, C#, JavaScript, R
- **Machine learning:** PyTorch, TensorFlow, Hugging Face, distributed and multi-GPU training
- **Computational biology:** NCBI, Bioconductor, Biopython, GWAS Catalog, PyMOL, ChimeraX, molecular dynamics (OpenMM), docking tools

Teaching Experience

- 2026 Senior Design Project II TA, Bilkent University, Mert Bıçakçı
- 2026 Algorithms and Programming II TA, Bilkent University, Prof. Uğur Güdükbay
- 2024 – 2025 Algorithms and Programming I TA, Bilkent University, Prof. Uğur Doğrusöz
- 2023 Advanced Programming TA, University of Tabriz, Dr. Sokhandan

Awards & Honours

- 2024 **Admission to fully funded M.Sc. in Computer Engineering**
Fully funded scholarship for the M.Sc. at Bilkent University (RA & TA)
- 2023 **Ranked in the top 10% of the Computer Engineering cohort**, University of Tabriz
- 2018 **Awarded a full scholarship for undergraduate study**
Ranked in the top 2% of more than 150,000 candidates in the nationwide university entrance examination (Konkur).

Software

2026 **RNP Bench** — a browser-based workbench for RNA–protein molecular dynamics: interface and physics analysis, system planning, ready-to-run inputs for **GROMACS**, **AMBER** and **OpenMM**, and trajectory analysis. Optional **OpenMM** backend for all-atom simulation on a CUDA GPU. 

Selected Projects

2026	Off-target gRNA identification in CRISPR systems. (Information Retrieval Systems Course)
2026	RNA–protein docking using diffusion models. (Deep Generative Models Course)
2026	Reinforcement learning agent defending genomic beacons against reconstruction attacks. (Data Privacy Course)
2025	Protein–drug interaction prediction with contrastive loss and RNN sequence encoders. (Machine Learning Course)
2024	Knowledge distillation in large language models. (Deep Learning Course)
2024	RNA sequence generation and optimisation using GANs and reinforcement learning. (Deep Learning Course)
2023	Semantic segmentation on CamVid: U-Net, PSP-Net and FPN in PyTorch.
2023	Generative models in PyTorch: DCGAN face synthesis (CelebA), CycleGAN translation, conditional GAN colourisation.
2023	DeepDream in PyTorch on a VGG16 backbone written from scratch and trained on CIFAR-10.

Academic Service

2026	Reviewer, <i>Bioinformatics</i> (Oxford University Press)
2026	Reviewer, RECOMB 2026
2026	Mentored an internship student, Çiçek Lab, Bilkent University